Q. N0:4

TCS Interview Question

Question description

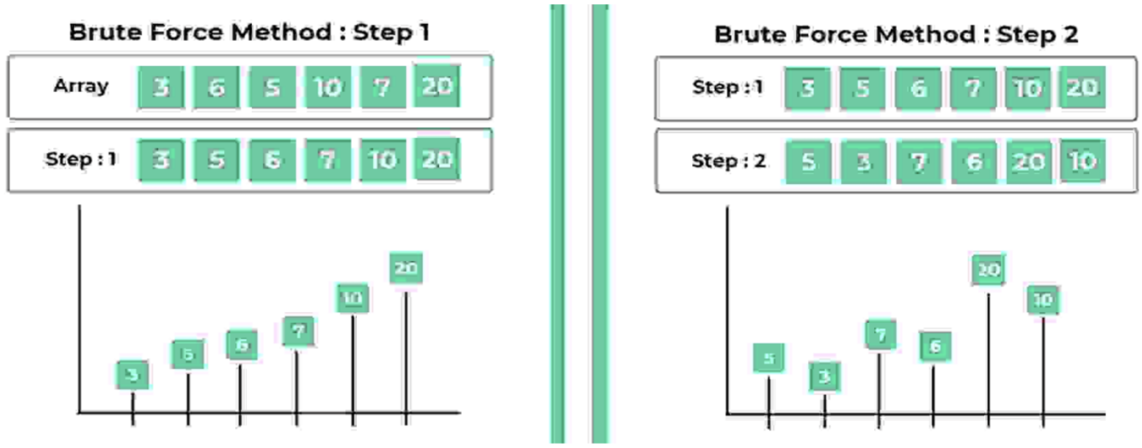
Ravi participated in TCS off campus interview at reputed institution, one of the technical question he has to complete with in the given time, where you need to sort the array in the waveform. There might be multiple possible output of the program. the following pattern output is appreciated.

Function Description

This is a simple method of solving this question which contains basic 2 steps and they are as follow

Step : 1 – Sort the array in ascending order.
Step : 2 – Swap all adjacent elements of the array

Let us consider the input array be {3, 6, 5, 10, 7, 20}. After sorting, we get {3, 5, 6, 7, 10, 20}. After swapping adjacent elements, we get {5, 3, 7, 6, 20, 10}.



Constraints

0<n< 100

0<arr<1000

Input Format
-First line contains the value of n that is the total number of elements in the array.
-Second line contains the space separated elements of array.

Output Format
-Output contains only line that is only the resultant array in the wave form fashion.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
6
3 6 5 10 7 20
```

EXPECTED OUTPUT

```
5 3 7 6 20 10
```

Test Case 2

INPUT (STDIN)

```
8
10 5 6 3 2 20 98 80
```

EXPECTED OUTPUT

```
3 2 6 5 20 10 98 80
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
int array[n]
```

Test Case 2

KEYWORD

```
for(int i=0;i<n;i++)
```

Test Case 3

KEYWORD

```
if(array[i]>array[j])
```

Complexity Test Cases

Code:

```
#include <stdio.h>

int main()
{
int n;
scanf("%d", &n);

int array[n];

for(int i = 0; i < n; i++)
{
scanf("%d", &array[i]);
}
```

```
for(int i = 0; i < n; i++)
{
for(int j = i + 1; j < n; j++)
{
if(array[i] > array[j])
{
int temp = array[i];
array[i] = array[j];
array[j] = temp;
}
}
}
for(int i = 0; i < n - 1; i += 2)
{
int temp = array[i];
array[i] = array[i + 1];
array[i + 1] = temp;
}
for(int i = 0; i < n; i++)
{
printf("%d", array[i]);

if(i < n - 1)
printf(" ");
}

return 0;
}
```

Output;

Test _case 1:

```
6
3 6 5 10 7 20
5 3 7 6 20 10%
aadityajaiswal@Aadityas-
```

Test _case 2:

```
8
10 5 6 3 2 20 98 80
3 2 6 5 20 10 98 80%
```